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(a)(i)	Electric field strength at a point is the electric force exerted per unit positive charge placed at that point.
(ii)	The electric field strength at a point is equal to the negative electric potential gradient at that point.
(b)(i)	2.9 cm
(ii)	At the maximum, the electric potential gradient is zero. Therefore, the resultant electric field strength is zero.
	$E_A = E_B$ $\frac{1}{4\pi\epsilon_0} \frac{q_A}{x^2} = \frac{1}{4\pi\epsilon_0} \frac{q_B}{(7.0 \times 10^{-2} - x)^2}$
	$q_B = \left(\frac{7.0 \times 10^{-2} - x}{x} \right)^2 q_A$ $= \left(\frac{7.0 \times 10^{-2} - 2.9 \times 10^{-2}}{2.9 \times 10^{-2}} \right)^2 (-2.0 \times 10^{-9})$ $= -4.0 \times 10^{-9} \text{ C}$
(iii)	The electron accelerates towards charge B until $x = 2.9 \text{ cm}$.
	After $x = 2.9 \text{ cm}$, it accelerates in the opposite direction towards charge A.
	The electron oscillates about $x = 2.9 \text{ cm}$.

Anglo-Chinese Junior College
H2 (9749) Physics

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